2.5.24

EE25BTECH11058 - Mohana Krishna Sushma

Question:
If $|\mathbf{a}| = 2$, $|\mathbf{b}| = 7$ and $\mathbf{a} \times \mathbf{b} = 3\hat{\imath} + 2\hat{\jmath} + 6\hat{k}$, find the angle between $\mathbf{a}$ and $\mathbf{b}$.
Solution:
Let the vector $\mathbf{c} = \mathbf{a} \times \mathbf{b}$. We are given:

$$\|\mathbf{a}\| = 2 \tag{1}$$
$$\|\mathbf{b}\| = 7 \tag{2}$$
$$\mathbf{c} = \begin{pmatrix} 3 \\ 2 \\ 6 \end{pmatrix} \tag{3}$$

First, we calculate the squared magnitude of $\mathbf{c}$ using the matrix product $\mathbf{c}^\top \mathbf{c}$:

$$\|\mathbf{c}\|^2 = \mathbf{c}^\top \mathbf{c} \tag{4}$$
$$= \begin{pmatrix} 3 & 2 & 6 \end{pmatrix} \begin{pmatrix} 3 \\ 2 \\ 6 \end{pmatrix} \tag{5}$$
$$= (3)(3) + (2)(2) + (6)(6) \tag{6}$$
$$= 9 + 4 + 36 \tag{7}$$
$$= 49 \tag{8}$$

Therefore, the magnitude of $\mathbf{c}$ is $\|\mathbf{c}\| = \sqrt{49} = 7$.
Using Lagrange's identity ,

$$\|\mathbf{a} \times \mathbf{b}\|^2 + (\mathbf{a}^\top \mathbf{b})^2 = \|\mathbf{a}\|^2 \|\mathbf{b}\|^2 \tag{9}$$

Substituting the magnitudes we have:

$$\|\mathbf{c}\|^2 + (\mathbf{a}^\top \mathbf{b})^2 = (2)^2 (7)^2 \tag{10}$$
$$49 + (\mathbf{a}^\top \mathbf{b})^2 = (4)(49) \tag{11}$$
$$49 + (\mathbf{a}^\top \mathbf{b})^2 = 196 \tag{12}$$

Solving for the squared dot product:

$$(\mathbf{a}^\top \mathbf{b})^2 = 196 - 49 = 147 \tag{13}$$

$$\mathbf{a}^\top \mathbf{b} = \sqrt{147} = \sqrt{49 \times 3} = 7\sqrt{3} \tag{14}$$

The definition of the dot product in terms of the angle θ between the vectors:

$$\mathbf{a}^\top \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos(\theta) \tag{15}$$

Substituting the known values:

$$7\sqrt{3} = (2)(7) \cos(\theta) \tag{16}$$
$$7\sqrt{3} = 14 \cos(\theta) \tag{17}$$

Solving for $\cos(\theta)$:

$$\cos(\theta) = \frac{7\sqrt{3}}{14} = \frac{\sqrt{3}}{2} \tag{18}$$

This implies that the angle θ is:

$$\theta = \frac{\pi}{6} \text{ or } 30^\circ \tag{19}$$